Rotaract Club of Xiamen University Malaysia | Rotary Club of Kuala Lumpur Dynamics | Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club
Submission Deadline: 25 October 2026

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

https://warcorb-demo.vercel.app/https://warcorb-demo.vercel.app/

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

SDC HACKATHON 2026

Technical & Tools Declaration

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1. Team & Project Overview

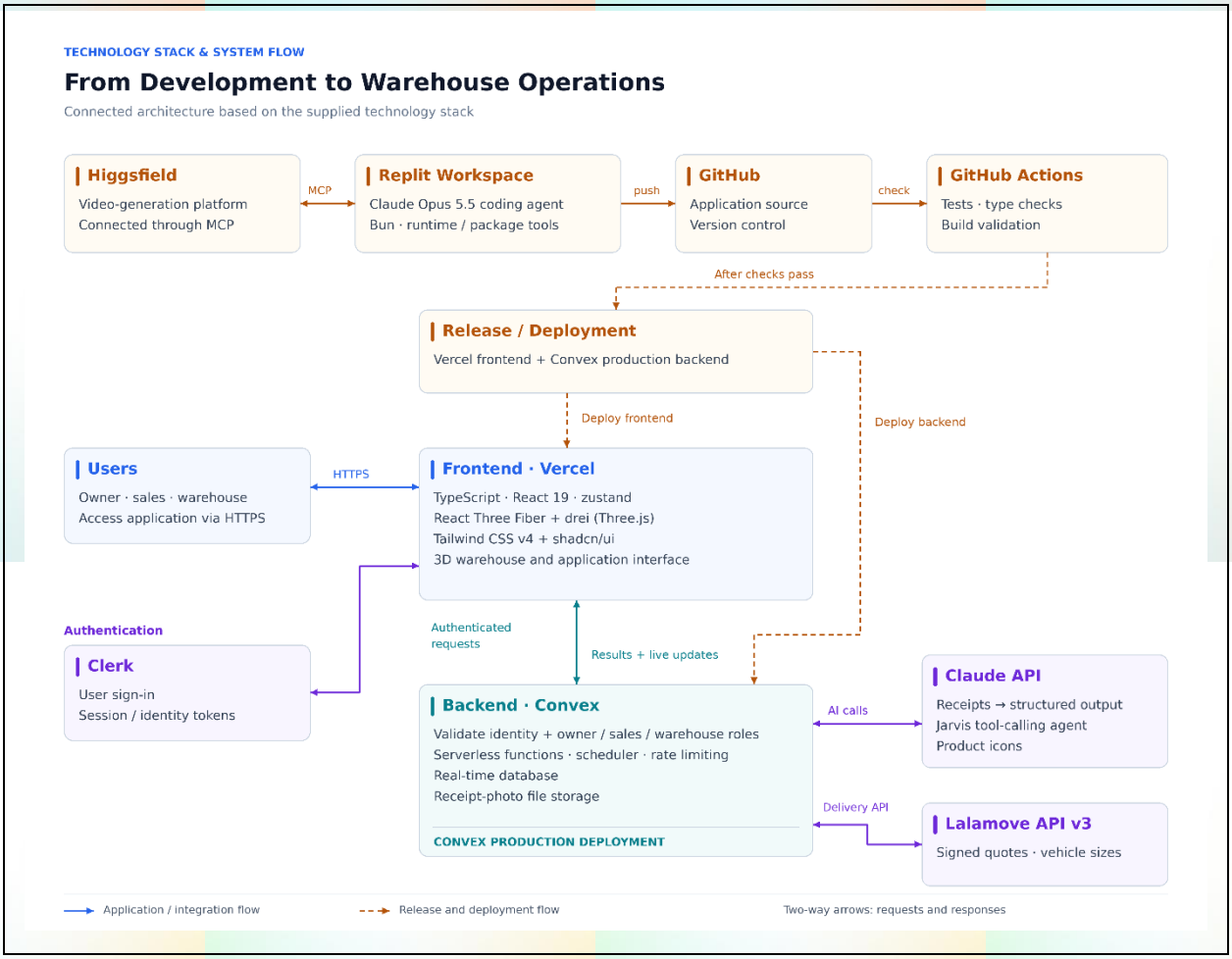
Team Name	Bababoi
Project / Solution Name	WareOrbs
Name (Leader)	Alfred Lee Khai Zher
Email & Phone (Leader)	alfredleekz@gmail.com / 0175657674
University Affiliation(s)	Taylor's University
Live Prototype URL	https://wareorb-demo.vercel.app/
Code Repository URL (public / access granted to judges)	https://github.com/POTATO0826/Bababoi.git

2. Architecture & Technology Stack

2.1 System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]



2.2 Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
TypeScript 5.9.3 (run with Bun 1.4.3)	Main programming language for frontend and backend. Bun is the runtime, dev server, bundler and test runner.	Apache-2.0 (TypeScript), MIT (Bun)
React 19.3.1 + React DOM 19.3.1	User interface framework for the web app	MIT
Three.js 0.180.0 + React Three Fiber 9.8.2 + drei 10.7.9	3D isometric warehouse map: furniture, racks, camera and selection	MIT
Convex 1.48.0 (+ @convex-dev/rate-limiter 0.4.2)	Backend framework: real-time database, server functions, file storage, scheduler, rate limiting	Apache-2.0
Anthropic SDK (@anthropic-ai/sdk) 0.131.0	Calls the Claude API for receipt reading and the Jarvis AI agent	Apache-2.0
Tailwind CSS 4.3.3 + shadcn/ui (@shadcn/react 0.3.1) + Radix UI 1.8.1	Styling and ready-made UI components (dialogs, buttons, chat panel)	MIT

3. APIs & Cloud Services

3.1 External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API & Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
Replit (Replit Agent, Claude Opus 5.5)	Coding agent used to build WareOrb. Connects to Higgsfield through the Higgsfield MCP server for video generation.	Sponsor-provided
Higgsfield API, via Higgsfield MCP (Higgsfield AI)	AI video generation, driven from Replit's Opus 5.5 agent	Sponsor-provided (through Replit) *or* paid, see check below
Claude API (Anthropic)	Reads delivery-note photos into draft receipts (structured output). Powers the Jarvis AI agent's tool calls. Generates product icons.	Paid (pay-as-you-go API key)
Lalamove API v3 (Lalamove)	Gets delivery quotes and vehicle sizes from the buyer's address and the cargo size and weight	Free (developer sandbox)

3.2 Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Convex Cloud (Convex)	Hosted backend and real-time database: stores SKUs, stock, receipts, deals, quotes and the audit log; runs server functions, file storage and scheduled jobs	Free (Starter plan)

Service & Provider	Purpose in Project	Plan / Region
Vercel (Vercel)	Hosts the web app frontend over HTTPS. Each push rebuilds the site and deploys the Convex backend with it.	Free (Hobby plan) / Global edge network
Clerk (Clerk)	Authentication: staff sign-in and verified email for owner, sales and warehouse roles	Free plan / Global
GitHub + GitHub Actions (GitHub / Microsoft)	Stores the source code and automatically runs tests, type checks and the build on every push	Free / Global
Replit (Replit)	Cloud development workspace and compute where the team codes with the Claude Opus 5.5 agent and runs Higgsfield video generation through MCP	Sponsor-provided / Global

4. AI Tools, Models & Datasets

4.1 AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Claude Opus 5 (`claude-opus-5`)	Anthropic, via Claude API	Pre-trained. Reads delivery-note photos into draft receipts, runs the Jarvis agent's tool calls, generates product icons. If Claude declines a request, the API re-runs it on Anthropic's recommended fallback model.	Anthropic Commercial Terms of Service + Usage Policy
Claude Opus 5.5	Anthropic, via Replit Agent	Pre-trained. Coding agent used to build WareOrb; drives Higgsfield video generation through MCP.	Replit Terms of Service + Anthropic Usage Policy
Higgsfield video model *(add the model name)*	Higgsfield AI, via Higgsfield MCP in Replit	Pre-trained. AI video generation	Higgsfield Term

4.2 Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
Live business records (staff accounts, receipts, buyer names, delivery addresses, quotes, audit log)	Entered by the SME's staff in the running app; stored in Convex Cloud	Private	Owned by the business using WareOrb	**Y**: staff names and emails, buyer contact and address. Access is limited by sign-in and role checks, and nothing is used for training.

4.3 AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	**Used For**	**Extent of Use (minor / moderate / major)**
Replit Agent (Claude Opus 5.5)	Writing and debugging application code: frontend, 3D warehouse map, Convex backend	Major
OpenAI Codex	Product and workflow planning documents; implementation and code review	Moderate
Claude Code / Claude (Anthropic)	Code changes and handoff docs; drafting this application form and the architecture diagram	Moderate
Higgsfield (via Higgsfield MCP in Replit)	AI-generated video content	Moderate
Replit Agent (Claude Opus 5.5)	Writing and debugging application code: frontend, 3D warehouse map, Convex backend	Major

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

Every AI suggestion was treated as a draft. A team member read each code change before merging. GitHub Actions ran automated tests, type checks and a build on every push, and failing changes were fixed or rejected. We wrote tests for the risky rules, such as unauthorised access, double posting and stock reservations, and walked through the full workflow for each role. Facts and figures in our documents were checked against cited sources. Inside the app, AI output is never final: staff confirm every AI-read receipt before stock changes. The team takes full responsibility for the submitted work.

5. Sponsor-Provided Resources

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Platform: Replit cloud workspace with Replit Agent (Claude Opus 5.5)	Main coding environment: the team wrote, debugged and tested WareOrb's frontend, 3D warehouse map and Convex backend with the Opus 5.5 agent
Replit	Tool integration: Higgsfield MCP connection inside Replit	Let the Opus 5.5 agent drive Higgsfield's video-generation platform to produce the project's video content

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Replit was essential to how we built the prototype, but not to how it runs. Its Claude Opus 5.5 agent let a small student team write, debug and test a full-stack app with a 3D warehouse map, a real-time backend and AI features within the competition timeline. Its Higgsfield connection produced our video content. The finished app runs independently on Convex and Vercel, so a shop using WareOrb does not depend on Replit. Another coding environment could replace it, but would have slowed development considerably.

Check before submitting:

- **Higgsfield as a sponsor:** if Higgsfield is itself an official SDC sponsor, change the second row's Sponsor to "Higgsfield" and the resource to "Credits / platform".
- **Section consistency:** make sure 3.1 (Access Tier), Section 3 of the proposal (sponsor answer) and this section all name the same sponsor or sponsors.

6. Security, Privacy & Responsible Use

6.1 Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

WareOrb stores business data: staff names and emails, buyer names and optional phone numbers, delivery addresses, delivery-note photos, voice recordings, stock records and an audit log.

Data sent to third parties:
- Clerk: staff sign-in details.
- Claude API: delivery-note images, and Jarvis chat text with the stock figures it needs.
- Lalamove: pickup and destination addresses and cargo details only, never buyer names or phone numbers.

Protection: every request requires sign-in, and the server checks the user's active staff role before returning or changing data. Deactivated staff lose access immediately. Traffic uses HTTPS, API keys stay in server settings and never reach the browser, and nothing is used to train models.

Retention: records are kept as the business's stock and audit history until the owner deletes them. The app does not yet delete old photos or recordings automatically.

6.2 Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

Hallucination: AI can misread handwriting or invent a SKU. Mitigation: AI output is only a draft. Extracted lines must match the catalogue, duplicate notes are rejected, and a worker confirms the physical count and bay before stock changes. Jarvis answers from live database tools, not memory, and cannot approve, post or collect.

Rate limits and cost: per-minute limits on Claude, Fish Audio and Lalamove calls stop runaway usage. Repeat quote requests reuse unexpired quotes.

Provider dependency: every AI feature has a fallback. Without Claude, Jarvis uses

keyword rules in English and Mandarin, and receipts are entered manually. Without Fish Audio, browser speech is used. Without Lalamove, verified manual carrier quotes are recorded.

Sandbox quotes: Lalamove sandbox prices are labelled and cannot be approved as real.

Bias: speech recognition may struggle with mixed Malay, English and Mandarin speech, so typed commands are always available.

7. Compliance & Readiness Checklist

Tick all boxes that apply before submitting.

- ☑ All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- ☑ All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- ☑ No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- ☑ Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- ☑ Any use of official SDC sponsor resources has been declared in Section 5.
- ☑ The prototype described here is accessible to judges through the links provided in Section 1.
- ☑ All fields in this template have been completed within the word limits stated.

8. Declaration

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

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24/10/2005s